93. The expression $\dfrac{\left(x+\frac{1}{y}\right)^a \cdot \left(x-\frac{1}{y}\right)^b}{\left(y+\frac{1}{x}\right)^a \cdot \left(y-\frac{1}{x}\right)^b}$ reduces to

(*a*) $\left(\frac{x}{y}\right)^{a-b}$ (*b*) $\left(\frac{y}{x}\right)^{a-b}$

(*c*) $\left(\frac{x}{y}\right)^{a+b}$ (*d*) $\left(\frac{y}{x}\right)^{a+b}$

94. The value of $\left(x^{\frac{b+c}{c-a}}\right)^{\frac{1}{a-b}} \cdot \left(x^{\frac{c+a}{a-b}}\right)^{\frac{1}{b-c}} \cdot \left(x^{\frac{a+b}{b-c}}\right)^{\frac{1}{c-a}}$ is

(*a*) 1 (*b*) *a*

(*c*) *b* (*d*) *c*

95. If $x^{\frac{1}{p}} = y^{\frac{1}{q}} = z^{\frac{1}{r}}$ and $xyz = 1$, then the value of $p + q + r$ would be **(M.B.A. 2008)**

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) a rational number

96. If $a^x = b^y = c^z$ and $b^2 = ac$, then *y* equals

(*a*) $\frac{xz}{x+z}$ (*b*) $\frac{xz}{2(x-z)}$

(*c*) $\frac{xz}{2(z-x)}$ (*d*) $\frac{2xz}{(x+z)}$

97. If $a^x = b$, $b^y = c$ and $c^z = a$, then the value of *xyz* is **(M.B.A., 2005; R.R.B., 2008)**

(*a*) 0 (*b*) 1

(*c*) $\frac{1}{abc}$ (*d*) *abc*

98. If $2^x = 4^y = 8^z$ and $\left(\frac{1}{2x}+\frac{1}{4y}+\frac{1}{6z}\right) = \frac{24}{7}$, then the value of *z* is

(*a*) $\frac{7}{16}$ (*b*) $\frac{7}{32}$

(*c*) $\frac{7}{48}$ (*d*) $\frac{7}{64}$

99. Suppose $4^a = 5$, $5^b = 6$, $6^c = 7$, $7^d = 8$, then the value of *abcd* is **(A.A.O. Exam, 2009)**

(*a*) 1 (*b*) $\frac{3}{2}$

(*c*) 2 (*d*) $\frac{5}{2}$

100. If $abc = 1$, then $\left(\frac{1}{1+a+b^{-1}}+\frac{1}{1+b+c^{-1}}+\frac{1}{1+c+a^{-1}}\right) = ?$ **(C.D.S. 2004)**

(*a*) 0 (*b*) 1

(*c*) $\frac{1}{ab}$ (*d*) *ab*

101. If *a*, *b*, *c* are real numbers, then the value of $\sqrt{a^{-1}b}\cdot\sqrt{b^{-1}c}\cdot\sqrt{c^{-1}a}$ is

(*a*) *abc* (*b*) $\sqrt{abc}$

(*c*) $\frac{1}{abc}$ (*d*) 1

102. If $3^{(x-y)} = 27$ and $3^{(x+y)} = 243$, then *x* is equal to **(R.R.B., 2003)**

(*a*) 0 (*b*) 2

(*c*) 4 (*d*) 6

103. If $x^y = y^x$, then $\left(\frac{x}{y}\right)^{\frac{x}{y}}$ is equal to **(M.C.A., 2005)**

(*a*) $x^{\frac{y}{x}}$ (*b*) $x^{\frac{x}{y}-1}$

(*c*) 1 (*d*) $x^{\frac{x}{y}}$

104. If $4^{x+y} = 1$ and $4^{x-y} = 4$, then the values of *x* and *y* respectively are

(*a*) $-\frac{1}{2}$ and $\frac{1}{2}$ (*b*) $-\frac{1}{2}$ and $-\frac{1}{2}$

(*c*) $\frac{1}{2}$ and $-\frac{1}{2}$ (*d*) $\frac{1}{2}$ and $\frac{1}{2}$

105. If $2^{2x-1} + 4^x = 2^{x-\frac{1}{2}} + 2^{x+\frac{1}{2}}$, then *x* equals

(*a*) $\frac{1}{2}$ (*b*) $\frac{2}{3}$

(*c*) $\frac{3}{2}$ (*d*) 1

106. If $3^{2x-y} = 3^{x+y} = \sqrt{27}$, the value of *y* is **(R.R.B., 2005)**

(*a*) $\frac{1}{2}$ (*b*) $\frac{1}{4}$

(*c*) $\frac{3}{2}$ (*d*) $\frac{3}{4}$

107. If $3^x = 5^y = 45^z$, then **(C.D.S., 2002)**

(*a*) $\frac{2}{z}=\frac{1}{y}-\frac{1}{x}$ (*b*) $\frac{2}{y}=\frac{1}{x}-\frac{1}{z}$

(*c*) $\frac{2}{x}=\frac{1}{z}-\frac{1}{y}$ (*d*) $x + y + z = 0$

108. Given $2^x = 8^{y+1}$ and $9^y = 3^{x-9}$, the value of $x + y$ is **(M.B.A., 2010)**

(*a*) 18 (*b*) 21

(*c*) 24 (*d*) 27

109. What are the values of *x* and *y* that satisfy the equation $2^{0.7x} \cdot 3^{-1.25y} = \frac{8\sqrt{6}}{27}$? **(C.A.T., 2006)**